Sensitivity of AI-Generated Credit Ratings to Elicitation Choices

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Abstract

Large language models (LLMs) are increasingly prompted for credit assessments, yet it is unknown whether the model holds a stable latent view the prompt reads out, or the elicitation manufactures the rating. Using a pre-registered specification-curve design—a fixed 90-item battery with an objective Altman Z″ benchmark crossed with a factorial grid of elicitation choices (provider, model version, temperature, prompt paraphrase, output format, few-shot exemplars, presentation) across three model families (12,960 elicitations)—we measure how far a credit rating moves across defensible choices. The instability is not a single tunable knob: no axis explains more than ∼5% of rating variance, while run-to-run variation explains about a quarter. It survives honored determinism (permutation $\boldsymbol{p} = \mathbf{0.001}$). The opinion holds no stable resolution—chance-corrected agreement peaks at only $\boldsymbol{\kappa} = \mathbf{0.46}$, even at the coarse investment-grade/high-yield split, across which 27% of matched pairs disagree. Switching or pooling vendors does not eliminate it. Economically, the modal call turns over ∼57% of positions, and a post-hoc capital recast swings the same \$45bn book's Pillar-1 capital \$2.0–4.2bn. Instability also appears in 31 anonymized, perturbed real issuers benchmarked to disclosed agency ratings under a decontamination gate. The elicitation specification is itself a model-risk parameter.

Keywords: large language models, credit ratings, specification-curve analysis, model risk, reproducibility, investment grade

1 Introduction

Ask a large language model whether a firm is investment grade; then vary a defensible elicitation choice over identical financials—reword the prompt, change the output format, switch the temperature, or query a different vendor's model—and for 27% of matched pairs the verdict crosses the investment-grade/high-yield boundary—the single most consequential line in credit. That boundary gates index inclusion, ratings-based investment mandates, collateral eligibility, covenant triggers, and regulatory capital; a name that crosses it is a fallen angel, forced-sold by mandate-constrained holders. When the crossing is driven not by the firm's fundamentals but by how the rating was elicited—which model, which wording, which format—the model has stopped measuring credit and started manufacturing it.

A rapidly growing body of practice treats LLMs as credit, risk, and trading analysts [1]: desks prompt a model for a rating view, a buy/hold/sell call, or a directional signal, and act on the answer. The workflow rests on a tacit assumption of *elicitation*—that the model holds a stable latent assessment of the firm and the prompt merely reads it out. If that holds, prompt wording, output format, and vendor choice are administrative details. If it fails, they are confounds, and every downstream credit decision inherits their variance. That LLM outputs are prompt-sensitive is documented in computer science [2, 3] and, for LLM-driven stock recommendations, in finance [4]; what is not established is the magnitude of that sensitivity in the units a risk officer acts on, or whether its cause is fixable.

This paper measures the assumption directly and prices it. Borrowing the specification-curve method from empirical social science [5, 6] and, in a management-annotation setting, Camuffo et al.[7], we hold the credit question fixed and vary only defensible researcher choices, then observe how far the rating moves. Our contribution is fourfold. First, we quantify the sensitivity of machine credit ratings against an objective benchmark and separate a fixable cause ("pick better prompts") from an unfixable one ("the noise floor is intrinsic"). Second, we translate the instability into the quantities a desk acts on—migration across the IG/HY line, portfolio turnover, implied credit spread, and regulatory capital. Third, we show the fragility survives the natural rebuttals: honored determinism, and vendor choice. Fourth, we answer the "use real data" objection with a decontaminated real-firm arm: we show that naive real-issuer evaluation is contaminated by model memorization, build a perturbed, screened, and fingerprint-gated sample of real rated issuers, and show that substantial specification instability is observed there too. We make no claim that LLMs rate credit *accurately*; our estimand throughout is stability, which is what a risk manager needs before any accuracy claim can matter [8].

2 Related work

Prompt sensitivity and non-determinism of LLMs. Modern LLMs are transformer-based foundation models [9–11]. A growing literature documents that their outputs vary with semantically irrelevant features of the prompt—paraphrase, format, option order, and sampling—even when the underlying task is fixed [2, 12]. Mizrahi et al.[13] show single-prompt evaluation is fragile and call for multi-prompt reporting. In finance, Chon et al.[4] report multifaceted variability in LLM-driven stock recommendations. Our work differs in three ways: we use an objective benchmark rather than human labels, we cross a factorial grid of choices rather than perturbing one axis at a time, and we price the resulting dispersion in regulatory and market units.

Specification-curve and multiverse analysis. The idea that a single analytical pipeline is one of many defensible ones is the "garden of forking paths" [14]; undisclosed analytic flexibility can present anything as significant [15], and multiverse and specification-curve analysis report results across the defensible choices [5, 6]. Camuffo et al.[7] apply a related design to human management annotation. We import the method to machine elicitation, treating each prompt-and-model configuration as one specification and the LLM as the estimator under test.

LLMs in finance, credit, and supervision. General-purpose [16–18] and finance-specialized [19, 20] LLMs are now applied to financial-statement analysis [21], return prediction [22], central-bank communication [23], risk profiling [1], banking supervision [24], and credit-rating forecasting [8], which finds that traditional methods still outperform generative models on accuracy. These studies target capability; we target the prior question of reproducibility, which conditions any capability claim.

Rating economics and model risk. Statistical default prediction from financial ratios is long-established [25, 26]; so are the information carried by fine rating distinctions and the disagreement among human raters [27–29]. Rating migration across the IG/HY line has first-order pricing and capital consequences [30]. Supervisory model-risk guidance requires that model outputs be validated and their uncertainty characterized [31]; the Basel standardized approach maps external ratings to risk weights [32], and the Basel Committee has itself documented cross-bank dispersion in risk-weighted assets for identical portfolios [33]. We connect these threads by showing that, for a machine rater, the specification is a first-order source of that dispersion.

3 Results

Throughout, the grid crosses three providers at two snapshots each (a current and a prior version), except one prior-version xAI snapshot that returned empty and is retained as a documented dead cell, not imputed; cross-provider comparisons therefore rest on the two families at full snapshot coverage.

3.1 The instability is a noise floor, not a knob

A mixed-effects decomposition of the credit letter-rank attributes variance to the item, to each design axis, to the provider$\times$item interaction, and to residual seed-level noise (Table 1). The firm dominates ($\sim$63%), as it should. What matters for the practitioner is the rest: the researcher-choice axes are strikingly flat—no single axis exceeds five percent, answer presentation being largest at $\sim$5.0%, while the provider a desk would most expect to matter contributes only $\sim$1.3% as a main effect—yet residual seed-level run-to-run variation, the pure repeated-call draw, is the single largest non-item component at $\sim$25%. The instability is not concentrated in a design choice a careful analyst could optimize; it is dominated by residual run-to-run variation. A "better"

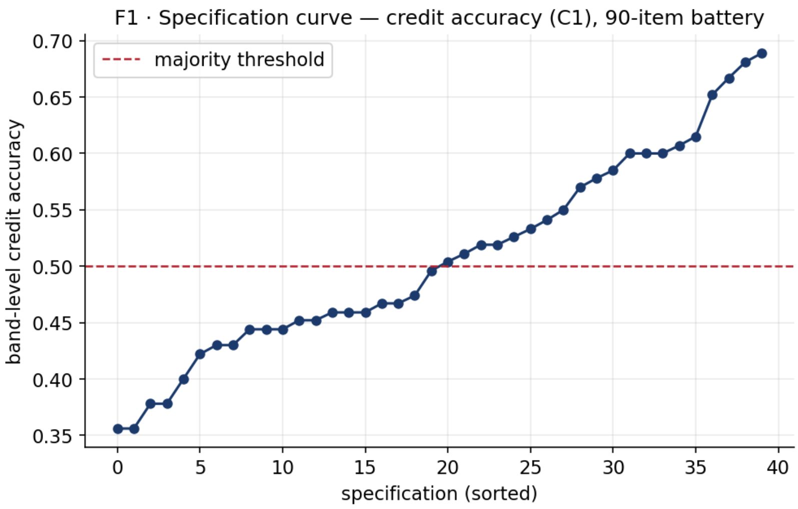

Fig. 1 Specification curve of the machine credit rating. Each point is one of the pre-registered specifications (three seeds pooled), ordered by majority-correct credit-band accuracy against the Altman Z″ benchmark. Accuracy ranges widely across specifications despite identical firm fundamentals, and roughly half reverse the pre-registered accuracy verdict (spec-curve flip share ≈ 0.50)—the multiverse the method is named for.

Table 1 Variance shares of the machine credit letter-rank (mixed-effects). Every controllable design choice is small relative to run-to-run noise.

Component	Variance share
Item (firm profile)	∼63%
Residual seed (run-to-run noise)	∼25%
Answer presentation (A_7, largest design axis)	∼5.0%
Provider × item interaction	∼2.2%
Model provider (A_1, main effect)	∼1.3%

prompt buys little, because the prompt is not where most of the movement lives: the specification curve of the credit rating (Fig. 1) fans out across defensible choices while the fixed firm fundamentals do not (Table 1).

3.2 Determinism does not save you

The natural rebuttal is that the noise floor is an artifact—two of the three providers ignore temperature = 0, so residual run-to-run variation might vanish under genuine determinism. We pre-registered the test. Restricting to the one provider×setting whose metadata confirms honored determinism (Google at temperature 0), the fragility does not vanish: across deterministic prompt configurations, majority-correct credit-band accuracy still ranges 0.36–0.69, and 62.5% of specifications flip the majority verdict, with a permutation test decisively rejecting specification-invariance (no permuted labeling exceeded the observed statistic; $p = (k+1)/(n+1) = 0.001$ over 999 permutations). Determinism removes run-to-run jitter but leaves specification fragility intact—the flip share inside the deterministic subgrid is higher, not lower, than the pooled estimate.

3.3 No vendor escape

The ∼1.3% provider main effect sits beside a fact that seems to contradict it. Across the two families at full snapshot coverage (OpenAI and Google; xAI contributes its current snapshot only, its prior snapshot being a documented dead cell), on identical firms across seeds, within-vendor agreement exceeds cross-vendor agreement at every granularity: 73.7% within versus 50.4% across at the band level, 22.5% versus 5.2% at the letter level. Both hold, and together they close both exits. Provider is a negligible *level* effect but enters through a large provider×item interaction: vendors disagree, but idiosyncratically, name by name, not as a constant shift a desk could calibrate out—unlike the persistent, characterizable differences of opinion documented among human rating agencies [27]. Switching vendors does not reliably change the rating on a given name, and pooling vendors does not eliminate the observed specification instability.

3.4 A rating that will not hold a notch—or the investment-grade line

Because the credit view is elicited at full letter resolution (AAA. . . C) and coarsened deterministically, we can ask the sharpest form of the question: at what rating granularity does the opinion become reproducible across specifications? It never fully does. Raw cross-specification agreement rises with coarseness but never clears 80%—16% at the letter grade, 34% at the notch, 65% at the three-way band, and 73% even at the two-way IG/HY split (equivalently, the disagreement shares plotted in Fig. 2). Chance-corrected, Fleiss' κ tops out at 0.46 (IG/HY) and falls to 0.39, 0.18, and 0.08 at finer scales; no granularity reaches the $\kappa = 0.6$ conventionally required for substantial agreement (Fig. 3). This is not a model hedging to one central grade: all 21 letter grades are used, with scale-usage entropy 4.03 of a possible 4.39 bits. The opinion is fully expressed and genuinely unstable, not compressed. The priced information that notch-level distinctions carry [28, 29] therefore cannot be reliably delivered by a machine rater elicited this way.

3.5 What it costs: turnover, spreads, and capital

The stability question becomes a markets question at the boundary. We define the per-comparison flip rate as follows: for each firm profile we draw two of its elicitations uniformly at random from the full grid—so that the two draws may differ on any elicitation axis (provider, model version, temperature, prompt paraphrase, output format, few-shot exemplars, answer presentation) and on the random seed, with no axis held fixed—and record whether they fall on opposite sides of the IG/HY line; the reported rate is the mean of this per-firm flip probability (one minus the Gini–Simpson concentration of the firm's IG/HY labels). On this measure, 26.8% (95% CI

Table 2 Fragility in decision and economic units.

Quantity	Result
Specifications flipping majority credit *band* verdict (pooled / determ.)	∼50% / 62.5%
Credit-band accuracy range, honored-determinism subgrid	0.36–0.69
Permutation test, determinism subgrid	$p = 0.001$
Within- vs cross-vendor agreement (band / letter)	73.7/50.4 · 22.5/5.2
Max chance-corrected agreement (Fleiss κ, IG/HY)	0.46 (no scale ≥ 0.6)
Per-comparison IG/HY boundary flip	26.8% (95% CI 20.9–32.5%)
Implied turnover (modal-call instability)	56.8% (95% CI 50.9–62.7%)
Implied spread range per name (non-saturating / median)	∼261 / ∼860 bps
Pillar-1 capital range, same \$45bn book (post-hoc cut)	\$2.0bn – \$4.2bn

20.9–32.5%) of matched elicitation pairs move a name across the IG/HY boundary. A desk mechanically following the modal call would incur 56.8% (95% CI 50.9–62.7%) turnover attributable purely to elicitation choices; the effect remains economically large even at the lower bound of each interval. Anchored to the Cornaggia et al. rating-spread schedule (80–140 bps per 2–3 notches) [30], the within-name spread range has a median near 860 bps, inflated by CCC saturation; for non-saturating names the median range is ∼261 bps, never below ∼106 bps. And in regulatory-capital units—mapping each specification's ratings through the Basel standardized approach [32] as a labeled post-hoc cut—the *same* \$45bn equal-weight corporate book requires between \$2.0bn and \$4.2bn of Pillar-1 minimum capital depending only on how the rating was requested. As a calibration (not an identity), two randomly chosen specifications differ on average by \$0.65bn, or 20.5% of the median—comparable to the ∼20% relative dispersion the Basel RCAP exercise found across internal-model banks holding identical portfolios [33]. The choice of *how* to ask, not *what* is asked, moves a name by hundreds of basis points and billions in capital across the most consequential boundary in credit (Table 2).

3.6 Real-firm robustness under a decontamination gate

A natural objection is that the instability reflects our constructed inputs. But evaluating LLMs on real issuers is itself contaminated: asked to name the company behind an anonymized financial profile, Gemini-Flash identifies 22% of mid-cap issuers from the figures alone, with hits clustered and reasoning-traced. Naive real-firm evaluation can therefore conflate memorized issuer information with independent credit judgment. We nonetheless build a decontaminated arm: 31 US mid-cap (\$2–15B) business-to-business issuers whose latest 10-K discloses a current agency rating (extracted with a verbatim quotation, permanent SEC URL, and document hash). Each dollar figure is scaled by a per-issuer factor drawn log-uniform$[0.6, 1.7]$—which preserves the financial-ratio structure used by the benchmark while disrupting exact-figure lookup—and consumer-facing firms are screened out. A pre-registered fingerprinting gate certifies decontamination at 2.9% identification. Rating each firm across a frozen

Table 3 Real-firm arm: 31 decontaminated business-to-business issuers, agency-rating benchmark.

Quantity	Result
Fingerprint identification, naive (exact figures)	22%
Fingerprint identification, decontaminated (gate)	2.9%
WATCH-stratum flip-share, real arm	0.226 (0.147–0.308)
WATCH-stratum flip-share, battery (same 12 specs)	0.318 (0.223–0.406)
Difference (real − battery), 95% CI	−0.091 (−0.213, 0.032)
Equivalence test (TOST, ±0.15 margin)	$p = 0.179$ (not established; underpowered)
Interpretation	substantial instability persists; equivalence not established
Machine band vs. disclosed agency rating (descriptive)	54%

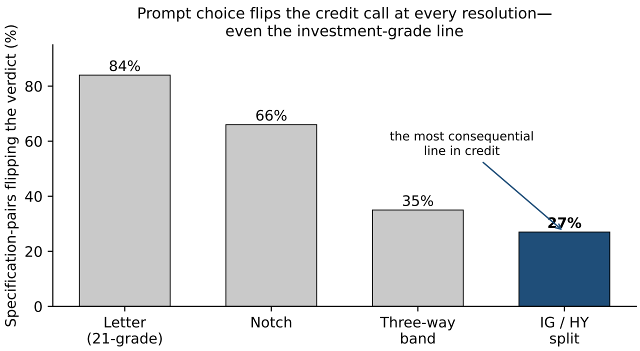

Fig. 2 Share of matched elicitation pairs whose credit verdicts differ, by rating granularity, where the two elicitations of a firm may differ on any grid axis (provider, model version, temperature, paraphrase, format, few-shot, presentation) and seed. Even at the coarse investment-grade/high-yield split—the most consequential line in credit—27% of matched pairs disagree.

12-specification grid at three seeds, and re-scoring the constructed battery on the identical grid, the real-arm WATCH flip-share is 0.226 (95% CI 0.147–0.308), compared with 0.318 (95% CI 0.223–0.406) in the matched battery. The estimated difference is −0.091 (95% CI −0.213 to 0.032) (Table 3). Although the difference is not statistically distinguishable from zero, formal equivalence within the pre-registered ±0.15 margin is not established (two-one-sided-tests $p = 0.179$), reflecting limited power in the 16-real/15-battery WATCH samples. The real-firm arm therefore provides evidence that substantial specification instability persists in real issuers, while the data are insufficient to establish quantitative equivalence with the constructed battery; the somewhat lower point estimate is consistent with real WATCH names sitting farther from band thresholds, not with residual memorization. Model bands match the disclosed agency rating only 54% of the time.

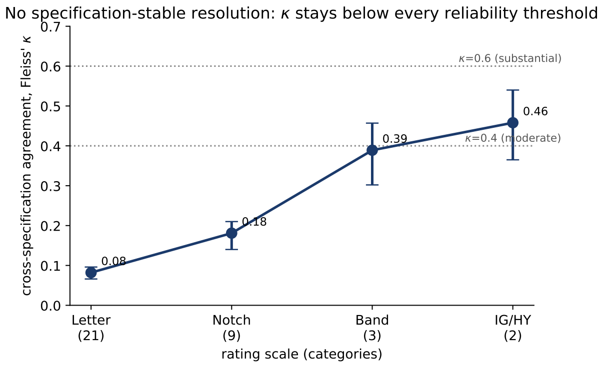

Fig. 3 Chance-corrected cross-specification agreement (Fleiss κ) by granularity; agreement never reaches the $\kappa = 0.6$ substantial-agreement threshold, peaking at 0.46 on the IG/HY split.

4 Discussion

The message for AI-assisted credit is blunt: a single-prompt LLM rating should not be treated as a specification-invariant measurement, it is one draw from a distribution wide enough to reclassify a firm across the investment-grade line, to churn a portfolio, and to double a capital charge. The fragility is dominated by residual run-to-run variation rather than any single elicitation choice a desk could optimize; it survives honored determinism; it cannot be escaped by switching or pooling vendors; and the machine credit opinion holds no stable rating resolution, not even the coarse IG/HY split. Critically, the pattern is not an artifact of synthetic inputs: the qualitative finding of substantial specification instability extends to real issuers under a design that first *demonstrates* the contamination of naive real-firm evaluation and then removes it. For a risk function, the practical implication is that the prompt specification is itself a model-risk parameter under any serious governance standard [31]: a rating reported without its specification curve overstates what the model actually determines. The fingerprinting result is a contribution in its own right—to our knowledge the first direct evidence that anonymized real-firm LLM evaluation can be contaminated by memorization even without names in the prompt—and it argues for decontamination gates as standard practice in financial LLM benchmarking.

4.1 Limitations

We make no capability claim: whether these models rate credit *well* is a separate question we do not adjudicate [8]. The finding is bounded to the three model families and settings studied, and one prior-version model contributed no data (a documented dead cell). Flip-share point estimates, though at full power, remain sample statistics on a 90-item battery; the real-firm arm comprises 31 issuers, thinned to a business-to-business, mid-cap universe by the decontamination screen, so its strata are smaller than the constructed battery's. The capital and spread translations are labeled post-hoc calibrations, not pre-registered confirmatory tests. Within those bounds, the stability conclusion is robust.

4.2 Future work

Natural extensions include larger and sector-stratified real-firm samples, retrieval-augmented and tool-using rating agents, human-in-the-loop workflows that report specification curves rather than point ratings, and decontamination gates applied to other high-stakes financial LLM tasks.

5 Conclusion

Reproducibility in AI-assisted credit cannot be assumed from a single prompt. Across a pre-registered specification curve, although firm fundamentals explain most rating variation, substantial specification- and run-level instability remains: the machine credit rating crosses the investment-grade line for 27% of matched elicitation pairs; it holds no stable resolution; and the instability survives determinism, vendor choice, and—under a decontamination gate—the move to real issuers. The specification is part of the result. A machine rating should be reported, governed, and priced with its specification curve attached.

6 Methods

Design rationale. We use a constructed battery as the primary identification environment because the estimand is elicitation stability, not performance on a naturally occurring issuer distribution. The battery holds firm fundamentals fixed, provides a pre-computable objective benchmark, and permits balanced coverage of credit-health strata while varying only the elicitation specification. Using real issuers alone would additionally risk contamination from model memorization; we therefore treat the decontaminated real-firm arm as a complementary external-validity test rather than the primary identification sample.

Battery and benchmark. The battery is a fixed set of 90 firm-quarter profiles, each carrying a compact financial statement (revenue, EBIT, total assets and liabilities, equity, working-capital components, retained earnings). The 90 profiles comprise two equally sized task families: 45 *credit-health* items, on which the model returns a letter/band credit rating, and 45 *directional* items, on which it returns a buy/hold/sell call (used for the turnover translation in Section 3.5). The credit-rating results and the IG/HY flip rate are computed on the 45 credit-health items; the directional items enter only the portfolio-turnover measure. For the credit task, each item's benchmark band is the Altman Z″ bond-rating equivalent [25] of its profile (coefficients 6.56/3.26/6.72/1.05; cutoffs 2.60/1.10), with the 45 credit-health items balanced 15/15/15 across the safe/watch/distress strata. The benchmark was frozen under a pre-registration hash before any model query; an independent post-hoc re-execution from the frozen facts reproduces all 45 credit-item bands.

Specification grid. Seven researcher-choice axes are crossed: provider (OpenAI, Google, xAI), model version (a current and a prior snapshot per provider), temperature (0 and 1.0), prompt paraphrase (three semantically equivalent templates), output format (JSON/free-text), few-shot exemplars (zero or two), and answer-option presentation (prose/original vs. table/reversed). A balance-constrained D-optimal fraction of 48 specifications was generated (main-effects model, effect-coded; seed frozen), with a post-hoc parsing-rule axis applied to stored text. Each specification is replicated across three seeds; the effective temperature returned in provider metadata is logged, because requested temperature = 0 is not uniformly honored.

Collection. The full design is 90 items × 48 specifications × 3 seeds = 12,960 elicitations; 10,793 returned content. One prior-version xAI snapshot returned empty for all 2,160 of its cells and is retained as a documented dead cell, not imputed. Each call is stored as an immutable JSON record; the raw corpus and every downstream stage are fixed by SHA-256 manifests. Collection is separated from analysis: the confirmatory analysis reads only the frozen response panel and never re-queries a model.

Parsing and statistics. Stored completions are parsed under three rules (strict, lenient, tolerant) applied post-hoc; the letter grade is mapped to notch, three-way band, and IG/HY coarsenings. Agreement is measured by Fleiss' κ; the contrast between homogeneous and heterogeneous ensembles by $\Delta\kappa$ with issuer-clustered bootstrap confidence intervals; specification-invariance by a label permutation test within the honored-determinism subgrid; and variance shares by a mixed-effects decomposition of the letter-rank with random effects for item, design axes, and their interactions.

Statistical analysis. All analyses are pre-registered unless labelled post-hoc, and inference uses distribution-free resampling throughout, so no normality assumption is made. Uncertainty on every headline quantity (the per-comparison IG/HY flip rate, $n = 45$ credit-health issuers; implied turnover, $n = 45$ directional issuers; and $\Delta\kappa$) is a percentile bootstrap resampled over issuer clusters (2,000 replicates for the economic quantities, 1,000 for the specification-curve quantities), and we report the point estimate with its two-sided 95% confidence interval. Specification-invariance is tested with a one-sided label permutation test on the between-specification spread of band accuracy in the honored-determinism subgrid (999 permutations; $p = (k+1)/(n+1)$), the direction being one-sided because only an excess of observed over permuted spread is evidence against invariance. Equivalence of the real-firm and battery WATCH flip-shares is assessed by two one-sided tests (TOST) against a pre-registered ± 0.15 margin ($n = 16$ real and 15 battery WATCH issuers). The significance threshold is $\alpha = 0.05$ for all tests. Because a specification curve evaluates many analytic paths, we guard against selective reporting by pre-registering the estimand and reporting the full curve rather than a chosen subset; the permutation test is itself a joint (multiplicity-aware) test of the whole grid. We follow the convention of reserving "significant" for results accompanied by a p-value and use "substantial" or "considerable" otherwise.

Economic translation (post-hoc). Rating dispersion is priced by anchoring to the Cornaggia et al. rating-spread schedule and to a cached daily option-adjusted-spread series; portfolio turnover is the share of names whose modal call changes across specifications. Regulatory capital is computed by mapping each specification's ratings through the Basel CRE20 external-ratings risk-weight table (quote-verified) on a \$45bn equal-weight book; the RCAP comparison is a calibration, not an identity. These translations are labeled post-hoc extensions of the frozen analysis.

Real-firm arm. We identify 31 US mid-cap (\$2–15B) business-to-business issuers whose most recent 10-K discloses a current long-term agency rating (S&P primary, else Moody's/Fitch mapped), extracted with a verbatim quotation, permanent SEC URL, and document hash. To preclude memorization, every dollar figure is multiplied by a single per-issuer factor drawn log-uniform$[0.6, 1.7]$, which preserves the financial-ratio structure used by the benchmark while disrupting exact-figure lookup; consumer-facing sectors are screened out. A pre-registered fingerprinting gate probes whether either model can name the issuer from the anonymized profile; the sample passes at 2.9% identification ($< 5\%$ threshold). Each firm is rated across a frozen 12-specification grid (presentation $\times$ provider $\times$ paraphrase $\times$ format) at three seeds, and the constructed battery is re-scored on the identical grid for a matched comparison on the WATCH stratum.

Reproducibility. The frozen battery, specification grid, response corpus (12,960 records), analysis code, and manifest hashes are archived; every confirmatory result regenerates byte-for-byte from the raw corpus through a single offline entry point, with no model calls.

Use of generative AI. Generative AI was used to assist with grammar correction and minor language edits of the authors' own text, and with software scaffolding, development, and coding of the analysis and reproducibility pipeline (Anthropic Claude), in all cases under the authors' direction. Separately, large language models are the object of study in this work—the credit-rating elicitations under analysis—as described above; every such elicitation is recorded in the specification grid and frozen corpus. The authors reviewed and verified all such output and take full responsibility for the content.

Data availability. All data and code required to reproduce the study are openly available in a deterministic, offline reproducibility capsule (CodeOcean DOI assigned at deposit; MIT license), mirrored in the source repository at https://github.com/samirrc2/prompt-choice-credit-ratings. Within the capsule, the `data/frozen/main/` directory contains the pre-registered 90-item firm battery (`battery_90.json`; 45 credit-health + 45 directional items), the specification grid (`grid_definition.json`), the prompt paraphrases (`paraphrase_templates.json`), the rating scale (`rating_scale.json`), and the Basel capital map (`capital_map.json`); `data/raw/main/` holds the complete frozen corpus of 12,960 model responses; `data/panel/panel.parquet` is the analysis panel built once from that corpus; and `data/frozen/realarm/` with `data/raw/realarm/` contain the decontaminated real-firm arm (the anonymized perturbed battery `battery_real45.json`, the sealed issuer crosswalk, the rating provenance `ratings_provenance.csv` with a verbatim 10-K quotation, permanent SEC URL, and document hash per issuer, and the arm/comparator/fingerprint model-output panels). The analysis code is in `analysis/`, pre-computed outputs and figures in `results/`, and SHA-256 integrity manifests in `manifest/`. A single offline entry point, `reproduce.sh`, regenerates every reported number, table, and figure byte-for-byte, with no model calls and no API keys. Issuer identities in the real-firm arm are anonymized and the sealed crosswalk is withheld; the underlying SEC filings (Form 10-K and company-facts XBRL) are publicly available through the SEC EDGAR system, with permanent document URLs listed in `data/frozen/realarm/ratings_provenance.csv`.

Code availability. All collection, panel-construction, and analysis code is released under the MIT license in the same repository; a single `reproduce.sh` regenerates every reported number, table, and figure from the frozen data offline. Provider API keys are required only to re-collect the corpus (not to reproduce any result) and are not distributed.

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Ethics declarations. This study did not involve human participants, human tissue, or animal subjects, and used no personally identifiable or sensitive personal data; ethical approval was therefore not required. The analysis uses only synthetic firm profiles, publicly available corporate financial disclosures (SEC filings), and the authors' own model elicitations.

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